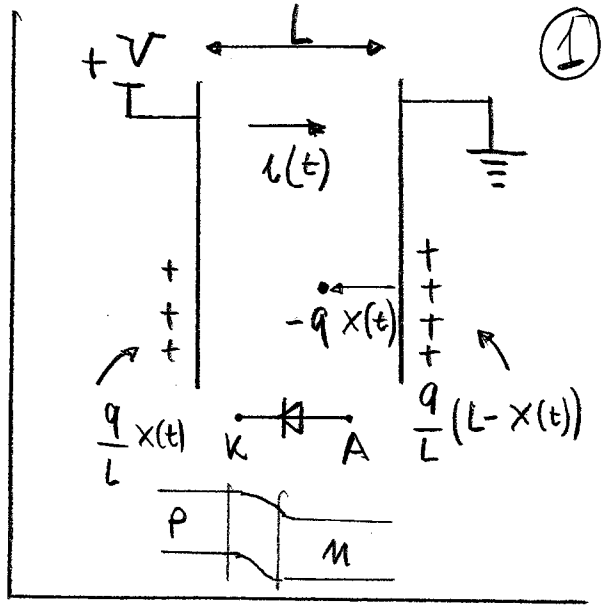
Shot noise



Reverse current in a reverse biased fotodiode

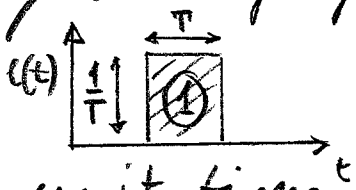
Because of EM induction a neg. charge $-q$ flowing from A to K produces the charges $\oplus Q = q/L\, x(t)$ in K e $q - Q = \frac{q}{L}(L - x(t))$ in A. The current is

$$i(t) = \frac{dQ}{dt} = q/L\, V(t) \leftarrow \text{Speed of } q \quad V \simeq 10^6 - 10^7\, m/s$$

V constant: $V = L/T$ [T = transit time]

The current is given by the superposition of random pulses

$$i(t) = [\vartheta(t) - \vartheta(t-T)]\frac{q}{T}$$

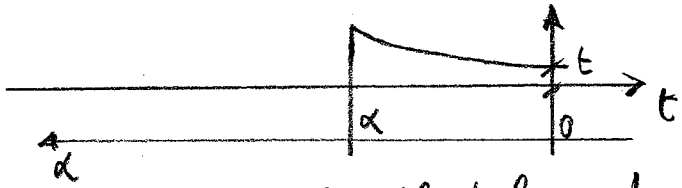


$$i(t) = q\, h(t) \rightarrow h(t) = \frac{1}{T}[\vartheta(t) - \vartheta(t-T)]$$

$\boxed{n}$ average pulses per unit time

Let's change $h(t) \rightarrow h(t) = \vartheta(t) \cdot 1/\tau \exp(-t/\tau)$ $\quad \int_0^\infty h(t)\,dt = 1$

$i(t) = q\, h(t)$



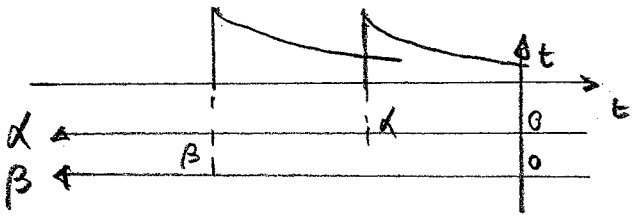
Reverse time axis α

Average current: A pulse that has started at α (delay) gives a contribution to the current in t given by $q \cdot h(\alpha)$. Probability $\alpha \rightarrow \alpha + d\alpha$

$$I = \langle i(t) \rangle = \int_0^\infty q\, h(\alpha) \langle n \rangle\, d\alpha = q \langle n \rangle \int_0^\infty h(\alpha)\, d\alpha = q \langle n \rangle \quad \left[P_{\alpha \rightarrow \alpha + d\alpha} = \langle n \rangle\, d\alpha \right.$$

$I = q \langle n \rangle$ as expected

We need to calculate $\langle i^2(t)\rangle$ Mean square value of the current $\rightarrow$ 2 pulses



For 2 pulses that have started exactly at α & β : (2)
the contribution to the square current in t is:

$$[q\,h(\alpha) + q\,h(\beta)]^2 = q^2 h^2(\alpha) + q^2 h^2(\beta) + q^2 h(\alpha)\,h(\beta) + q^2 h(\beta)\,h(\alpha)$$

Probability that 1 pulse has started at α OR β : $P_{\alpha\to\alpha+d\alpha} = \langle n\rangle d\alpha$, $P_{\beta\to\beta+d\beta} = \langle n\rangle d\beta$

Probability that 1 pulse has started at α AND 1 at β : $P_{\substack{\alpha\to d\alpha\\ \beta\to d\beta}} = \langle n\rangle^2 d\alpha\, d\beta$

$$\langle i^2(t)\rangle = \int_0^\infty q^2 h^2(\alpha)\langle n\rangle d\alpha + \int_0^\infty\int_0^\infty q^2 h(\alpha)\,h(\beta)\langle n\rangle^2 d\alpha\, d\beta$$

$$\langle i^2(t)\rangle = q^2\langle n\rangle\int_0^\infty h^2(\alpha)\,d\alpha + q^2\langle n\rangle^2 \underbrace{\int_0^\infty h(\alpha)\,d\alpha}_{1} + \underbrace{\int_0^\infty h(\beta)\,d\beta}_{1}$$

$$\langle i^2(t)\rangle = q^2\langle n\rangle\int_0^\infty h^2(\alpha)\,d\alpha + \langle i(t)\rangle^2$$

In general for a quantity $X \rightarrow \sigma_X^2 = \langle X^2(t)\rangle - \langle X(t)\rangle^2$

$$\Rightarrow \sigma_n^2 = \langle i^2(t)\rangle - \langle i(t)\rangle^2 = q\langle i(t)\rangle\cdot\int_0^\infty h^2(\alpha)\,d\alpha$$

Note that the variance of $i(t)$ does not depend on $\langle i(t)\rangle$ $\leftarrow$ deterministic

<u>Spectral analysis:</u>

$$\sigma_n^2 = \langle P_n\rangle = \int_0^\infty S(f)\,df = q\langle i(t)\rangle\int_0^\infty h^2(\alpha)\,d\alpha$$

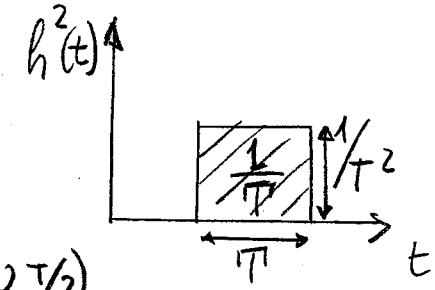
$$\int_0^\infty S(f)\,df = 2q\langle i(t)\rangle\int_0^\infty |H(\omega)|^2\frac{d\omega}{2\pi} \quad \circledast$$

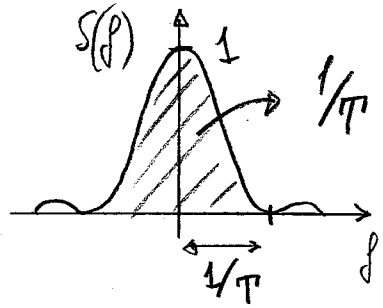
<u>Parseval Theorem:</u>

$$\int_0^\infty |h(t)|^2 = \int_{-\infty}^{+\infty}|H(\omega)|^2\frac{d\omega}{2\pi} = 2\int_0^{+\infty}|H(\omega)|^2\frac{d\omega}{2\pi}$$

Let's come back to the original pulse shape $\rightarrow$ $h(t) = [\vartheta(t) - \vartheta(t-T)]\,1/T$

Since the integrals (*) are extended over the whole spectral range we can remove them





$$S(f) = 2q\langle i(t)\rangle\,|H(\omega)|^2$$

$$\mathcal{F}\left[\mathrm{Rect}\left(\frac{t}{T}\right)\right] = \mathcal{F}\left[\vartheta(t-T/2)-\vartheta(t+T/2)\right]\frac{1}{T} = \frac{\sin(\omega T/2)}{\omega T/2}$$

$$|H(\omega)|^2 = \mathrm{sinc}^2\left(\frac{\omega T}{2}\right) = \mathrm{sinc}^2(\pi f T)$$

Max $|H(\omega)|^2 = 1$ 1st zero $|H(\omega)| \rightarrow \pi f T = \pi \Rightarrow$ 1st zero $= 1/T$

Therefore:

$$S(0) = 2q\langle i(t)\rangle \text{ and } \Delta f = 1/T \quad \left[T = L/v = \frac{10^{-5}\,m}{10^{7}\,m/s} \simeq 10^{-12}\,s\right]$$

The bandwidth of shot noise is $\Delta f \simeq 1\,THz \rightarrow$ White noise

$$\Rightarrow S(f) = S(0) = 2q\langle i(t)\rangle \quad \forall f$$

$$\langle i^2_{shot}(t)\rangle = \int_0^{\Delta f} S(f)\,df = 2q\langle i(t)\rangle\,\Delta f$$ ← For a detector with bandwidth Δf

$$i^{RMS}_{shot} = \sqrt{\langle i^2_{shot}(t)\rangle} = \sqrt{2q\langle i(t)\rangle\,\Delta f}$$

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